```python
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.impute import KNNImputer, SimpleImputer
from sklearn.preprocessing import OneHotEncoder, OrdinalEncoder,
LabelEncoder, MinMaxScaler
from sklearn.compose import ColumnTransformer
from sklearn.naive_bayes import GaussianNB, CategoricalNB
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import classification_report
from mixed_naive_bayes import MixedNB
from sklearn.linear_model import LogisticRegression

df = pd.read_csv("../J1/HeartDiseaseUCI.csv", index_col=0)

num_variables = ["age", "trestbps", "chol", "thalach", "ca",
"oldpeak"]
cat_variables = ["cp", "restecg", "thal"]
ord_variables = ["slope"]
bin_variables = ["sex", "fbs", "exang"]

df["target"] = np.where(df["num"] >= 1, 1, 0)

X = df.drop(["target", "num"], axis=1)
y = df["target"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size = 0.2,
    stratify = df["target"],
    random_state = 314
)

print(X_train.shape)
print(X_test.shape)
print(y_train.shape)
print(y_test.shape)
```

```
(242, 13)
(61, 13)
(242,)
(61,)
```

4. Logistic Regression

```python
num_pipeline = Pipeline(steps = [
    ("imputer", KNNImputer()),
    ("scaler", MinMaxScaler())
])
```

```
cat_pipeline = Pipeline(steps = [
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("encoder", OneHotEncoder(drop = "first"))
])
ord_pipeline = Pipeline(steps = [
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("encoder", OrdinalEncoder())
])
bin_pipeline = Pipeline(steps = [
    ("imputer", SimpleImputer(strategy="most_frequent")),
])

preprocessor = ColumnTransformer(transformers=
     [
        ("num", num_pipeline, num_variables),
        ("cat", cat_pipeline, cat_variables),
        ("ord", ord_pipeline, ord_variables),
        ("bin", bin_pipeline, bin_variables),
     ],
 )

pipeline = Pipeline(steps = [
    ("preprocesor", preprocessor),
    ("logistic", LogisticRegression())
])

end_C = 1
step_C = 0.02
start_C = step_C

hyperparameters = [
    {
        "logistic__penalty" : ["l1", "l2", "elasticnet"],
        "logistic__C"       : np.arange(start_C, end_C, step_C),
        "logistic__max_iter": [10000],
        "logistic__solver"  : ["lbfgs", "liblinear", "newton-cg",
"newton-cholesky", "sag", "saga"]
    },
    {
        "logistic__penalty" : [None],
        "logistic__max_iter": [10000],
        "logistic__solver"  : ["lbfgs", "liblinear", "newton-cg",
"newton-cholesky", "sag", "saga"]
    }
]

gscv_logistic = GridSearchCV(
    estimator = pipeline,
    param_grid=hyperparameters,
    cv = 5,
```

```
    scoring = "f1",
    verbose = 4,
)

gscv_logistic.fit(X_train, y_train)
```

```
Fitting 5 folds for each of 888 candidates, totalling 4440 fits
[CV 1/5] END logistic__C=0.02, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.02, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.02, logistic__max_iter=10000,
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0.0s
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0.0s
[CV 5/5] END logistic__C=0.02, logistic__max_iter=10000,
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0.0s
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time=   0.1s
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```

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[CV 3/5] END logistic__C=0.02, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.783 total time=
0.0s
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logistic__penalty=l2, logistic__solver=lbfgs;, score=0.750 total time=
0.0s
[CV 5/5] END logistic__C=0.02, logistic__max_iter=10000,
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total time=   0.0s
```

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time=   0.0s
```

```
[CV 4/5] END logistic__C=0.02, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.0s
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logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.02, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.02, logistic__max_iter=10000,
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```

```
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time=   0.0s
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[CV 3/5] END logistic__C=0.04, logistic__max_iter=10000,
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[CV 4/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.000 total time=
0.1s
[CV 5/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.000 total time=
0.1s
[CV 1/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.789 total time=
0.1s

[CV 3/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.766 total time=
0.0s
[CV 4/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.791 total time=
0.0s
[CV 5/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.791 total time=
0.0s
[CV 1/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.933 total
time=   0.0s
[CV 2/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.792 total
time=   0.1s
[CV 4/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.723 total
time=   0.1s
[CV 5/5] END logistic__C=0.04, logistic__max_iter=10000,
```

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logistic__penalty=l2, logistic__solver=liblinear;, score=0.756 total
time=   0.0s
[CV 1/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.905 total
time=   0.1s
[CV 2/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.789 total
time=   0.1s
[CV 3/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.766 total
time=   0.0s
[CV 4/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.791 total
time=   0.0s
[CV 5/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.791 total
time=   0.0s
[CV 1/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.905
total time=   0.0s
[CV 2/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.789
total time=   0.1s
[CV 3/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.766
total time=   0.0s
[CV 4/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.791
total time=   0.1s
[CV 5/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.791
total time=   0.0s
[CV 1/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.789 total time=
0.0s
[CV 3/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.766 total time=
0.0s
[CV 4/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.791 total time=
0.0s
[CV 5/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.791 total time=
0.0s
[CV 1/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.905 total time=
```

```
0.0s
[CV 2/5] END logistic__C=0.04, logistic__max_iter=10000,
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0.0s
[CV 3/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.766 total time=
0.1s
[CV 4/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.791 total time=
0.1s
[CV 5/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.791 total time=
0.1s
[CV 1/5] END logistic__C=0.04, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.1s
[CV 5/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.1s
[CV 2/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.1s
[CV 3/5] END logistic__C=0.04, logistic__max_iter=10000,
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total time=   0.0s
[CV 4/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
```

```
[CV 3/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.04, logistic__max_iter=10000,
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time=   0.0s
[CV 2/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.04, logistic__max_iter=10000,
```

```
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.04, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s

[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.732 total
time=   0.0s
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[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.830 total
time=   0.1s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.731 total
time=   0.1s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.720 total
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[CV 1/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
```

```
time=   0.0s
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logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
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logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.732 total time=
0.0s
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0.0s
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0.1s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.649 total time=
0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.595 total time=
0.0s
[CV 1/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.905 total time=
0.1s
```

```
[CV 2/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.789 total time=
0.0s
[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.766 total time=
0.0s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.791 total time=
0.1s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.791 total time=
0.0s
[CV 1/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.933 total
time=   0.0s
[CV 2/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.789 total
time=   0.0s
[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.792 total
time=   0.0s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.739 total
time=   0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.773 total
time=   0.0s
[CV 1/5] END logistic__C=0.06, logistic__max_iter=10000,
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time=   0.0s
[CV 2/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.789 total
time=   0.0s
[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.766 total
time=   0.0s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.791 total
time=   0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.791 total
time=   0.0s
[CV 1/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.905
total time=   0.0s
[CV 2/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.789
total time=   0.0s
[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
```

```
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.766
total time=   0.0s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.791
total time=   0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.791
total time=   0.0s
[CV 1/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.789 total time=
0.0s
[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.766 total time=
0.0s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.791 total time=
0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.791 total time=
0.0s
[CV 1/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.789 total time=
0.1s
[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.766 total time=
0.0s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.791 total time=
0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.791 total time=
0.1s
[CV 1/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
```

```
time=   0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s

[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
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logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
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logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
```

```
[CV 1/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.06, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.732 total
time=   0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
```

```
logistic__penalty=l1, logistic__solver=liblinear;, score=0.766 total
time=   0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.857 total
time=   0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.723 total
time=   0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.756 total
time=   0.0s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.1s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
```

```
0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.732 total time=
0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.722 total time=
0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.780 total time=
0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.615 total time=
0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.667 total time=
0.0s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.789 total time=
0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.791 total time=
0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.791 total time=
0.0s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.909 total
time=   0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.789 total
time=   0.0s

[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.792 total
time=   0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.773 total
time=   0.0s
```

```
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.773 total
time=   0.0s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.905 total
time=   0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.789 total
time=   0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.791 total
time=   0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.791 total
time=   0.0s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.905
total time=   0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.789
total time=   0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.792
total time=   0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.791
total time=   0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.791
total time=   0.0s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.789 total time=
0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.791 total time=
0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.791 total time=
0.0s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
```

```
logistic__penalty=l2, logistic__solver=saga;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
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0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.791 total time=
0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.791 total time=
0.0s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
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time=   0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
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total time=   0.0s
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total time=   0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
```

```
total time=   0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
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score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
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score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
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time=   0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
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time=   0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.08, logistic__max_iter=10000,
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time=   0.0s
[CV 2/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
```

```
[CV 4/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.08, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
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0.0s
[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
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time=   0.0s

[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
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time=   0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
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[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
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time=   0.0s
[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
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time=   0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
```

```
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total time=   0.0s
[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.1, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
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0.0s
[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.789 total time=
0.0s
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.810 total time=
0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.615 total time=
0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.714 total time=
0.0s
[CV 1/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.905 total time=
```

```
0.0s
[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
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0.0s
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.791 total time=
0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.791 total time=
0.1s
[CV 1/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.909 total
time=   0.0s
[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.0s
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.792 total
time=   0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.773 total
time=   0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.773 total
time=   0.0s
[CV 1/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.905 total
time=   0.0s
[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.789 total
time=   0.0s
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.791 total
time=   0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.791 total
time=   0.0s
[CV 1/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.905
total time=   0.0s
[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.789
total time=   0.0s
```

```
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.792
total time=   0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.791
total time=   0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.791
total time=   0.0s
[CV 1/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.789 total time=
0.0s
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.791 total time=
0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.791 total time=
0.0s
[CV 1/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
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0.0s
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.791 total time=
0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.791 total time=
0.0s
[CV 1/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
```

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logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s

[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
```

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score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
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[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.1, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.870 total time=   0.0s
```

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[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.810 total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.844 total time=   0.0s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.744 total time=   0.0s
[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.756 total time=   0.0s
[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
```

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logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.789 total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.844 total time=   0.0s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.683 total time=   0.0s
[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.744 total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.789 total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.792 total time=   0.0s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.791 total time=   0.0s

[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.791 total time=   0.0s
[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.909 total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.769 total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.792 total time=   0.1s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
```

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logistic__solver=liblinear;, score=0.773 total time=   0.0s
[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.773 total time=   0.0s
[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.789 total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.792 total time=   0.0s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.791 total time=   0.0s
[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.791 total time=   0.0s
[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
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[CV 4/5] END logistic__C=0.12000000000000001,
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[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.791 total time=   0.0s
[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.789 total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.792 total time=   0.0s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.791 total time=   0.0s
[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.791 total time=   0.0s
```

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[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.789 total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.792 total time=   0.1s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.791 total time=   0.0s
[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.791 total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.12000000000000001,
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logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
```

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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
```

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logistic__solver=saga;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s

[CV 5/5] END logistic__C=0.12000000000000001,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.1s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.909 total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.780 total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.864 total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.714 total time=   0.0s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.727 total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.1s
```

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[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.800 total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.870 total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.714 total time=   0.0s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.744 total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
```

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logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.789 total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.792 total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.791 total time=   0.0s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.818 total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.930 total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.769 total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.792 total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.773 total time=   0.0s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.773 total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.789 total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.792 total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.791 total time=   0.1s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.818 total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
```

```
logistic__solver=newton-cholesky;, score=0.789 total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.792 total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.791 total time=   0.0s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.818 total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.789 total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.792 total time=   0.0s

[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.791 total time=   0.0s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.818 total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.789 total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.792 total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.791 total time=   0.0s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.818 total time=   0.1s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
```

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[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.13999999999999999,
```

```
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.1s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.13999999999999999,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.909 total
```

```
time=   0.0s
[CV 2/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.780 total
time=   0.0s
[CV 3/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.837 total
time=   0.0s
[CV 4/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.732 total
time=   0.1s
[CV 5/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.756 total
time=   0.1s
[CV 1/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.1s

[CV 1/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
```

```
[CV 3/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.905 total time=
0.0s
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logistic__penalty=l2, logistic__solver=lbfgs;, score=0.791 total time=
0.0s
[CV 5/5] END logistic__C=0.16, logistic__max_iter=10000,
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time=   0.0s
[CV 3/5] END logistic__C=0.16, logistic__max_iter=10000,
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time=   0.0s
[CV 3/5] END logistic__C=0.16, logistic__max_iter=10000,
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time=   0.1s
[CV 4/5] END logistic__C=0.16, logistic__max_iter=10000,
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time=   0.1s
[CV 5/5] END logistic__C=0.16, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.0s
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total time=   0.1s
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total time=   0.1s
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total time=   0.0s
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```

```
0.0s
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total time=   0.0s
[CV 1/5] END logistic__C=0.16, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.0s
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total time=   0.0s
[CV 4/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.1s
```

```
[CV 5/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.18, logistic__max_iter=10000,
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time=   0.0s
[CV 2/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.1s
[CV 5/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.1s
[CV 5/5] END logistic__C=0.18, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.1s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.19999999999999998,
```

```
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.933 total time=   0.0s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.762 total time=   0.1s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.837 total time=   0.0s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.744 total time=   0.0s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.800 total time=   0.1s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.1s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.1s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
```

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score=nan total time=   0.1s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.909 total time=   0.1s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.769 total time=   0.0s

[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.870 total time=   0.0s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.714 total time=   0.1s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.818 total time=   0.0s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.789 total time=   0.1s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.792 total time=   0.0s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.810 total time=   0.0s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.818 total time=   0.0s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.930 total time=   0.0s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.769 total time=   0.1s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.792 total time=   0.0s
```

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[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.773 total time=   0.0s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.800 total time=   0.0s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.789 total time=   0.1s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.792 total time=   0.0s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.810 total time=   0.0s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.818 total time=   0.0s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.905 total time=   0.1s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.789 total time=   0.1s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.792 total time=   0.1s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.810 total time=   0.1s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.818 total time=   0.1s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.905 total time=   0.1s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.789 total time=   0.0s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.792 total time=   0.0s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.810 total time=   0.0s
[CV 5/5] END logistic__C=0.19999999999999998,
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```
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.818 total time=   0.0s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.789 total time=   0.0s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.792 total time=   0.0s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.810 total time=   0.0s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.818 total time=   0.0s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
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[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
```

```
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s

[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
```

```
[CV 3/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.19999999999999998,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.780 total
time=   0.0s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.884 total
time=   0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.744 total
time=   0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
```

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logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.1s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
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logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
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logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.909 total time=
0.0s
[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.870 total time=
0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.714 total time=
0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.818 total time=
```

```
0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.789 total time=
0.0s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.810 total time=
0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.930 total
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[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.0s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.792 total
time=   0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.773 total
time=   0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.905 total
time=   0.0s
[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.789 total
time=   0.1s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.810 total
time=   0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.905
total time=   0.1s
```

```
[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.789
total time=   0.0s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.792
total time=   0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.810
total time=   0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.0s

[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.789 total time=
0.1s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.810 total time=
0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.789 total time=
0.0s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
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0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.810 total time=
0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
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logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
```

```
score=nan total time=   0.1s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.22, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.1s
```

```
[CV 1/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.933 total
time=   0.0s
[CV 2/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.780 total
time=   0.0s
[CV 3/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.884 total
time=   0.0s
[CV 4/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.744 total
time=   0.0s
[CV 5/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.1s
[CV 5/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
```

```
[CV 2/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.1s
[CV 3/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.909 total time=
0.0s
[CV 2/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.870 total time=
0.0s
[CV 4/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.714 total time=
0.0s
[CV 5/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.789 total time=
0.0s
[CV 3/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.810 total time=
0.0s
[CV 5/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.930 total
time=   0.0s
[CV 2/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.0s
[CV 3/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.792 total
```

```
time=   0.0s
[CV 4/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.773 total
time=   0.1s
[CV 5/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.905 total
time=   0.1s
[CV 2/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.789 total
time=   0.1s
[CV 3/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.0s
[CV 4/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.810 total
time=   0.1s
[CV 5/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.0s
[CV 1/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.905
total time=   0.1s
[CV 2/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.789
total time=   0.1s
[CV 3/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.792
total time=   0.1s
[CV 4/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.810
total time=   0.1s
[CV 5/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.1s
[CV 1/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.789 total time=
0.1s
[CV 3/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.810 total time=
0.1s
[CV 5/5] END logistic__C=0.24, logistic__max_iter=10000,
```

```
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.789 total time=
0.1s
[CV 3/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.810 total time=
0.0s
[CV 5/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
0.1s
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[CV 2/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.1s
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total time=   0.0s
[CV 2/5] END logistic__C=0.24, logistic__max_iter=10000,
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[CV 3/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.1s
[CV 1/5] END logistic__C=0.24, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
```

```
[CV 2/5] END logistic__C=0.24, logistic__max_iter=10000,
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[CV 5/5] END logistic__C=0.26, logistic__max_iter=10000,
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time=   0.0s
[CV 3/5] END logistic__C=0.26, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.26, logistic__max_iter=10000,
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```
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total time=   0.0s
[CV 2/5] END logistic__C=0.26, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.26, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.0s
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total time=   0.1s
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logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.26, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.909 total time=
0.0s
[CV 2/5] END logistic__C=0.26, logistic__max_iter=10000,
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0.1s
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0.0s
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logistic__penalty=l1, logistic__solver=saga;, score=0.714 total time=
0.0s
[CV 5/5] END logistic__C=0.26, logistic__max_iter=10000,
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0.1s
```

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logistic__penalty=l2, logistic__solver=lbfgs;, score=0.789 total time=
0.0s
[CV 3/5] END logistic__C=0.26, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.26, logistic__max_iter=10000,
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0.0s
[CV 5/5] END logistic__C=0.26, logistic__max_iter=10000,
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0.0s
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time=   0.0s
[CV 3/5] END logistic__C=0.26, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.792 total
time=   0.0s
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time=   0.0s
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time=   0.1s
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time=   0.1s
[CV 3/5] END logistic__C=0.26, logistic__max_iter=10000,
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time=   0.0s
[CV 4/5] END logistic__C=0.26, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.810 total
time=   0.0s
[CV 5/5] END logistic__C=0.26, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.1s
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total time=   0.1s
[CV 4/5] END logistic__C=0.26, logistic__max_iter=10000,
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total time=   0.0s
[CV 5/5] END logistic__C=0.26, logistic__max_iter=10000,
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total time=   0.0s
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logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.26, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.810 total time=
0.1s

[CV 5/5] END logistic__C=0.26, logistic__max_iter=10000,
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```

```
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time=   0.0s
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total time=   0.0s
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score=nan total time=   0.0s
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```
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[CV 4/5] END logistic__C=0.28, logistic__max_iter=10000,
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[CV 5/5] END logistic__C=0.28, logistic__max_iter=10000,
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[CV 1/5] END logistic__C=0.28, logistic__max_iter=10000,
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time=   0.0s
[CV 2/5] END logistic__C=0.28, logistic__max_iter=10000,
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time=   0.0s
[CV 3/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.884 total
time=   0.0s
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total time=   0.0s
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logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
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logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
```

```
0.0s
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[CV 3/5] END logistic__C=0.28, logistic__max_iter=10000,
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0.0s

[CV 4/5] END logistic__C=0.28, logistic__max_iter=10000,
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0.0s
[CV 5/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.789 total time=
0.0s
[CV 3/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.809 total time=
0.1s
[CV 4/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.810 total time=
0.0s
[CV 5/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.930 total
time=   0.1s
[CV 2/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.0s
[CV 3/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.792 total
time=   0.0s
```

```
[CV 4/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.791 total
time=   0.0s
[CV 5/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.905 total
time=   0.0s
[CV 2/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.789 total
time=   0.0s
[CV 3/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.809 total
time=   0.0s
[CV 4/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.810 total
time=   0.0s
[CV 5/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.0s
[CV 1/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.905
total time=   0.0s
[CV 2/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.789
total time=   0.0s
[CV 3/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.809
total time=   0.0s
[CV 4/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.810
total time=   0.1s
[CV 5/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.0s
[CV 1/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.789 total time=
0.0s
[CV 3/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.809 total time=
0.0s
[CV 4/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.810 total time=
0.1s
[CV 5/5] END logistic__C=0.28, logistic__max_iter=10000,
```

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logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.789 total time=
0.0s
[CV 3/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.809 total time=
0.0s
[CV 4/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.810 total time=
0.1s
[CV 5/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
```

```
total time=   0.1s
[CV 2/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.1s
[CV 2/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.1s
[CV 1/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
```

```
[CV 3/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.28, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s

[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.1s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.930 total time=   0.0s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.780 total time=   0.0s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.884 total time=   0.1s
[CV 4/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.762 total time=   0.0s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.800 total time=   0.0s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.1s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.30000000000000004,
```

```
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.1s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.1s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
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[CV 4/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
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[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.1s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.909 total time=   0.0s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.769 total time=   0.0s
[CV 3/5] END logistic__C=0.30000000000000004,
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logistic__solver=saga;, score=0.851 total time=   0.1s
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logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.714 total time=   0.0s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
```

```
logistic__solver=saga;, score=0.818 total time=   0.0s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.905 total time=   0.1s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.769 total time=   0.0s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.809 total time=   0.0s
[CV 4/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.810 total time=   0.1s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.818 total time=   0.0s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.955 total time=   0.0s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.769 total time=   0.1s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.792 total time=   0.0s
[CV 4/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.791 total time=   0.0s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.800 total time=   0.0s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.769 total time=   0.0s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.809 total time=   0.1s
[CV 4/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.810 total time=   0.0s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.818 total time=   0.0s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.905 total time=   0.1s
```

```
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.769 total time=   0.0s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.809 total time=   0.0s
[CV 4/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.810 total time=   0.1s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.818 total time=   0.0s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.769 total time=   0.0s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.809 total time=   0.1s
[CV 4/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.810 total time=   0.0s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.818 total time=   0.0s

[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.905 total time=   0.1s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.769 total time=   0.0s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.809 total time=   0.1s
[CV 4/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.810 total time=   0.0s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.818 total time=   0.0s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.30000000000000004,
```

```
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.1s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.1s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
```

```
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.1s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.30000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.1s
```

```
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.930 total
time=   0.0s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.884 total
time=   0.0s
[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.762 total
time=   0.1s
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.1s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
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total time=   0.0s
[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
```

```
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s

[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.909 total time=
0.1s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
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0.0s
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0.1s
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logistic__penalty=l1, logistic__solver=saga;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.809 total time=
0.0s
[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.810 total time=
0.0s
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.955 total
time=   0.0s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.0s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.792 total
```

```
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logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.905 total
time=   0.1s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.809 total
time=   0.1s
[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.810 total
time=   0.1s
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.1s
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.905
total time=   0.1s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.769
total time=   0.0s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.809
total time=   0.0s
[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.810
total time=   0.1s
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.0s
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.809 total time=
0.0s
[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.810 total time=
0.1s
```

```
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.809 total time=
0.0s
[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.810 total time=
0.1s
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
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0.0s
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[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
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time=   0.0s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
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time=   0.0s
[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
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time=   0.0s
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.1s
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
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total time=   0.0s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
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total time=   0.0s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
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total time=   0.0s
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
```

```
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total time=   0.1s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
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total time=   0.0s
[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
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logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
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time=   0.0s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
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time=   0.0s

[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
```

```
time=   0.0s
[CV 3/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.32, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
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logistic__penalty=l1, logistic__solver=liblinear;, score=0.930 total
time=   0.0s
[CV 2/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 3/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.884 total
time=   0.0s
[CV 4/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.762 total
time=   0.0s
[CV 5/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
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time=   0.0s
[CV 2/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
```

```
[CV 4/5] END logistic__C=0.34, logistic__max_iter=10000,
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[CV 5/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
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logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.34, logistic__max_iter=10000,
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total time=   0.1s
[CV 3/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
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[CV 5/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
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[CV 4/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.909 total time=
0.0s
[CV 2/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.870 total time=
0.1s
[CV 4/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.714 total time=
0.1s
[CV 5/5] END logistic__C=0.34, logistic__max_iter=10000,
```

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logistic__penalty=l1, logistic__solver=saga;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.809 total time=
0.1s
[CV 4/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.810 total time=
0.1s
[CV 5/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.955 total
time=   0.1s
[CV 2/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.792 total
time=   0.1s
[CV 4/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.791 total
time=   0.1s
[CV 5/5] END logistic__C=0.34, logistic__max_iter=10000,
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time=   0.1s
[CV 1/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.905 total
time=   0.1s
[CV 2/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.809 total
time=   0.0s
[CV 4/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.810 total
time=   0.0s
[CV 5/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.1s
[CV 1/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.905
```

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total time=   0.0s
[CV 2/5] END logistic__C=0.34, logistic__max_iter=10000,
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total time=   0.0s
[CV 3/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.809
total time=   0.0s
[CV 4/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.810
total time=   0.0s
[CV 5/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.1s
[CV 1/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.809 total time=
0.0s
[CV 4/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.810 total time=
0.0s

[CV 5/5] END logistic__C=0.34, logistic__max_iter=10000,
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0.1s
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0.0s
[CV 2/5] END logistic__C=0.34, logistic__max_iter=10000,
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0.0s
[CV 3/5] END logistic__C=0.34, logistic__max_iter=10000,
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0.1s
[CV 4/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.810 total time=
0.1s
[CV 5/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
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[CV 2/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
```

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[CV 3/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.1s
[CV 1/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.34, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.0s
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total time=   0.0s
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total time=   0.0s
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total time=   0.1s
[CV 3/5] END logistic__C=0.34, logistic__max_iter=10000,
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total time=   0.1s
[CV 4/5] END logistic__C=0.34, logistic__max_iter=10000,
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total time=   0.0s
[CV 5/5] END logistic__C=0.34, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
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[CV 4/5] END logistic__C=0.34, logistic__max_iter=10000,
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score=nan total time=   0.1s
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[CV 2/5] END logistic__C=0.34, logistic__max_iter=10000,
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logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
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logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.1s
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logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
```

```
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[CV 2/5] END logistic__C=0.36000000000000004,
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logistic__solver=liblinear;, score=0.800 total time=   0.0s
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[CV 2/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.1s
[CV 4/5] END logistic__C=0.36000000000000004,
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[CV 4/5] END logistic__C=0.36000000000000004,
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[CV 2/5] END logistic__C=0.36000000000000004,
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[CV 5/5] END logistic__C=0.36000000000000004,
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[CV 2/5] END logistic__C=0.36000000000000004,
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logistic__solver=saga;, score=0.870 total time=   0.1s
[CV 4/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.714 total time=   0.0s
[CV 5/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.818 total time=   0.1s
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logistic__solver=lbfgs;, score=0.769 total time=   0.1s
[CV 3/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.809 total time=   0.0s
[CV 4/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.810 total time=   0.1s
[CV 5/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.818 total time=   0.1s
[CV 1/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.955 total time=   0.1s
[CV 2/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.769 total time=   0.1s
[CV 3/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
```

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logistic__solver=liblinear;, score=0.809 total time=   0.1s
[CV 4/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
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logistic__max_iter=10000, logistic__penalty=l2,
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[CV 1/5] END logistic__C=0.36000000000000004,
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[CV 4/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
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logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.818 total time=   0.1s
[CV 1/5] END logistic__C=0.36000000000000004,
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[CV 2/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
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[CV 3/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.809 total time=   0.0s
[CV 4/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.810 total time=   0.1s
[CV 5/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
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[CV 1/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
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[CV 2/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
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[CV 3/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.809 total time=   0.0s
[CV 4/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.810 total time=   0.0s
```

```
[CV 5/5] END logistic__C=0.36000000000000004,
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[CV 4/5] END logistic__C=0.36000000000000004,
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.36000000000000004,
```

```
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.1s
[CV 1/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.1s
[CV 4/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s

[CV 1/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.1s
[CV 2/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.1s
[CV 4/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
```

```
logistic__solver=saga;, score=nan total time=   0.1s
[CV 3/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.36000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.1s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.909 total
time=   0.1s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.884 total
time=   0.1s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.762 total
time=   0.1s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.783 total
time=   0.1s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
```

```
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.1s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.1s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.1s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.909 total time=
0.0s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.870 total time=
0.0s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.714 total time=
0.1s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
```

```
logistic__penalty=l1, logistic__solver=saga;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.809 total time=
0.1s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.810 total time=
0.0s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.955 total
time=   0.0s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.809 total
time=   0.0s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.791 total
time=   0.1s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.905 total
time=   0.0s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.809 total
time=   0.0s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.810 total
time=   0.0s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.0s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.905
```

```
total time=   0.0s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.769
total time=   0.0s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.809
total time=   0.1s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.810
total time=   0.0s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.0s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.905 total time=
0.0s

[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.809 total time=
0.0s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.810 total time=
0.0s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.809 total time=
0.0s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.810 total time=
0.0s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
```

```
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.1s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.1s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.1s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
```

```
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.1s
[CV 2/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.38, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.1s
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
```

```
0.1s
[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.909 total
time=   0.0s
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.884 total
time=   0.1s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.762 total
time=   0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.1s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
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total time=   0.1s
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s

[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.1s
```

```
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.909 total time=
0.1s
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.870 total time=
0.1s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.714 total time=
0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.810 total time=
0.1s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.955 total
time=   0.1s
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.0s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
```

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logistic__penalty=l2, logistic__solver=liblinear;, score=0.809 total
time=   0.1s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.791 total
time=   0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.905 total
time=   0.0s
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
time=   0.0s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.0s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.810 total
time=   0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.0s
[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.905
total time=   0.0s
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.769
total time=   0.0s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.792
total time=   0.0s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.810
total time=   0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.1s
[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.810 total time=
```

```
0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.810 total time=
0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
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time=   0.0s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
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[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.1s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
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time=   0.0s

[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.1s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.1s
[CV 1/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.4, logistic__max_iter=10000,
```

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logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.1s
[CV 5/5] END logistic__C=0.4, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
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logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.1s
[CV 4/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.909 total time=   0.1s
[CV 2/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.800 total time=   0.1s
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logistic__solver=liblinear;, score=0.800 total time=   0.1s
[CV 1/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.1s
[CV 2/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.42000000000000004,
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```

```
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logistic__solver=newton-cg;, score=nan total time=   0.1s
[CV 5/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cg;, score=nan total time=   0.1s
[CV 1/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.1s
[CV 3/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
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[CV 2/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
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[CV 3/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
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[CV 4/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
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logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.909 total time=   0.0s
[CV 2/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.769 total time=   0.1s
[CV 3/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.870 total time=   0.1s
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logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.714 total time=   0.1s
```

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[CV 5/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.818 total time=   0.1s
[CV 1/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.905 total time=   0.1s
[CV 2/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.769 total time=   0.1s
[CV 3/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.792 total time=   0.1s
[CV 4/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.810 total time=   0.1s
[CV 5/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.818 total time=   0.1s
[CV 1/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.955 total time=   0.1s
[CV 2/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.769 total time=   0.1s
[CV 3/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.809 total time=   0.1s
[CV 4/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.791 total time=   0.1s
[CV 5/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.800 total time=   0.1s
[CV 1/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.905 total time=   0.1s
[CV 2/5] END logistic__C=0.42000000000000004,
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logistic__solver=newton-cg;, score=0.769 total time=   0.1s
[CV 3/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.792 total time=   0.1s
[CV 4/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.810 total time=   0.0s
[CV 5/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.818 total time=   0.1s
[CV 1/5] END logistic__C=0.42000000000000004,
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```
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.905 total time=   0.1s
[CV 2/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.769 total time=   0.0s
[CV 3/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.792 total time=   0.1s
```

```
[CV 4/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.810 total time=   0.1s
[CV 5/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.818 total time=   0.1s
[CV 1/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.905 total time=   0.1s
[CV 2/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.769 total time=   0.0s
[CV 3/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.792 total time=   0.1s
[CV 4/5] END logistic__C=0.42000000000000004,
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score=0.810 total time=   0.0s
[CV 5/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.818 total time=   0.1s
[CV 1/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.905 total time=   0.1s
[CV 2/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
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[CV 3/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.792 total time=   0.1s
[CV 4/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.810 total time=   0.1s
[CV 5/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.818 total time=   0.0s
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logistic__solver=lbfgs;, score=nan total time=   0.1s
[CV 2/5] END logistic__C=0.42000000000000004,
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```

```
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logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.42000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.1s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
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[CV 1/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.909 total time=
0.1s
[CV 2/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.870 total time=
0.1s
[CV 4/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.744 total time=
```

```
0.1s
[CV 5/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.810 total time=
0.1s
[CV 5/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.955 total
time=   0.1s
[CV 2/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.809 total
time=   0.0s
[CV 4/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.791 total
time=   0.1s
[CV 5/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 1/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.905 total
time=   0.1s
[CV 2/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
time=   0.0s
[CV 3/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.1s
[CV 4/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.810 total
time=   0.1s
[CV 5/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.0s
```

```
[CV 1/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.905
total time=   0.1s
[CV 2/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.769
total time=   0.1s
[CV 3/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.792
total time=   0.0s
[CV 4/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.810
total time=   0.1s
[CV 5/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.0s

[CV 1/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.810 total time=
0.0s
[CV 5/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.810 total time=
0.1s
[CV 5/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.1s
[CV 2/5] END logistic__C=0.46, logistic__max_iter=10000,
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logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.1s
[CV 5/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.1s
[CV 1/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.1s
[CV 5/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.1s
[CV 3/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
```

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score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.46, logistic__max_iter=10000,
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[CV 5/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.1s
[CV 3/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.1s
[CV 3/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.46, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.1s
[CV 3/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=lbfgs;, score=nan total time=   0.0s
```

```
[CV 5/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
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[CV 1/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.909 total time=   0.1s
[CV 2/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.800 total time=   0.0s
[CV 3/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.909 total time=   0.0s
[CV 4/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.762 total time=   0.1s
[CV 5/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=liblinear;, score=0.800 total time=   0.0s
[CV 1/5] END logistic__C=0.48000000000000004,
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logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.48000000000000004,
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logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.48000000000000004,
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[CV 4/5] END logistic__C=0.48000000000000004,
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[CV 5/5] END logistic__C=0.48000000000000004,
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logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.48000000000000004,
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logistic__solver=newton-cholesky;, score=nan total time=   0.0s

[CV 3/5] END logistic__C=0.48000000000000004,
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logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
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[CV 1/5] END logistic__C=0.48000000000000004,
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[CV 3/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1, logistic__solver=sag;,
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[CV 1/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.909 total time=   0.0s
[CV 2/5] END logistic__C=0.48000000000000004,
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logistic__solver=saga;, score=0.769 total time=   0.0s
[CV 3/5] END logistic__C=0.48000000000000004,
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[CV 4/5] END logistic__C=0.48000000000000004,
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logistic__solver=saga;, score=0.762 total time=   0.0s
[CV 5/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l1,
logistic__solver=saga;, score=0.818 total time=   0.1s
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logistic__solver=lbfgs;, score=0.905 total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.769 total time=   0.0s
[CV 3/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.792 total time=   0.0s
[CV 4/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.810 total time=   0.1s
[CV 5/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=lbfgs;, score=0.818 total time=   0.1s
[CV 1/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
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[CV 2/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
```

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logistic__solver=liblinear;, score=0.769 total time=   0.1s
[CV 3/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.809 total time=   0.1s
[CV 4/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.791 total time=   0.0s
[CV 5/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=liblinear;, score=0.800 total time=   0.1s
[CV 1/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.905 total time=   0.1s
[CV 2/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.769 total time=   0.0s
[CV 3/5] END logistic__C=0.48000000000000004,
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[CV 4/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.810 total time=   0.0s
[CV 5/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cg;, score=0.818 total time=   0.0s
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[CV 2/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
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[CV 3/5] END logistic__C=0.48000000000000004,
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[CV 4/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.810 total time=   0.0s
[CV 5/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=newton-cholesky;, score=0.818 total time=   0.0s
[CV 1/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.769 total time=   0.0s
[CV 3/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.792 total time=   0.0s
```

```
[CV 4/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.810 total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=l2, logistic__solver=sag;,
score=0.818 total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.905 total time=   0.0s
[CV 2/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.769 total time=   0.1s
[CV 3/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.792 total time=   0.1s
[CV 4/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.810 total time=   0.0s
[CV 5/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=l2,
logistic__solver=saga;, score=0.818 total time=   0.0s
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logistic__solver=lbfgs;, score=nan total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.1s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=lbfgs;, score=nan total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=liblinear;, score=nan total time=   0.0s
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logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
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[CV 5/5] END logistic__C=0.48000000000000004,
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```
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logistic__solver=liblinear;, score=nan total time=   0.1s
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logistic__solver=newton-cg;, score=nan total time=   0.0s
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logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cg;, score=nan total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s

[CV 3/5] END logistic__C=0.48000000000000004,
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logistic__solver=newton-cholesky;, score=nan total time=   0.1s
[CV 4/5] END logistic__C=0.48000000000000004,
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[CV 5/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=newton-cholesky;, score=nan total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.1s
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=sag;, score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
```

```
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logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.1s
[CV 4/5] END logistic__C=0.48000000000000004,
logistic__max_iter=10000, logistic__penalty=elasticnet,
logistic__solver=saga;, score=nan total time=   0.0s
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[CV 1/5] END logistic__C=0.5, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.5, logistic__max_iter=10000,
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[CV 4/5] END logistic__C=0.5, logistic__max_iter=10000,
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0.0s
[CV 5/5] END logistic__C=0.5, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.5, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.909 total
time=   0.0s
[CV 2/5] END logistic__C=0.5, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 3/5] END logistic__C=0.5, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.884 total
time=   0.0s
[CV 4/5] END logistic__C=0.5, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.762 total
time=   0.1s
[CV 5/5] END logistic__C=0.5, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.5, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.5, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
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```
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logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
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logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
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logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
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total time=   0.1s
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logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.5, logistic__max_iter=10000,
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0.0s
[CV 2/5] END logistic__C=0.5, logistic__max_iter=10000,
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[CV 4/5] END logistic__C=0.5, logistic__max_iter=10000,
```

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0.0s
[CV 1/5] END logistic__C=0.5, logistic__max_iter=10000,
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0.0s
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0.0s
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0.0s
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[CV 4/5] END logistic__C=0.5, logistic__max_iter=10000,
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[CV 3/5] END logistic__C=0.5, logistic__max_iter=10000,
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[CV 4/5] END logistic__C=0.5, logistic__max_iter=10000,
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[CV 5/5] END logistic__C=0.5, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
```

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total time=   0.1s
[CV 3/5] END logistic__C=0.5, logistic__max_iter=10000,
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total time=   0.0s

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total time=   0.1s
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total time=   0.1s
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score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.5, logistic__max_iter=10000,
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[CV 4/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
```

```
0.0s
[CV 5/5] END logistic__C=0.52, logistic__max_iter=10000,
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[CV 1/5] END logistic__C=0.52, logistic__max_iter=10000,
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time=   0.1s
[CV 2/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 3/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.884 total
time=   0.1s
[CV 4/5] END logistic__C=0.52, logistic__max_iter=10000,
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time=   0.0s
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total time=   0.0s
```

```
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[CV 1/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.909 total time=
0.1s
[CV 2/5] END logistic__C=0.52, logistic__max_iter=10000,
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0.1s
[CV 3/5] END logistic__C=0.52, logistic__max_iter=10000,
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0.0s
[CV 4/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.762 total time=
0.0s
[CV 5/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.0s
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time=   0.0s
[CV 4/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.762 total
time=   0.1s
[CV 5/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.52, logistic__max_iter=10000,
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time=   0.1s
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time=   0.1s
[CV 3/5] END logistic__C=0.52, logistic__max_iter=10000,
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time=   0.0s
[CV 4/5] END logistic__C=0.52, logistic__max_iter=10000,
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time=   0.1s
[CV 5/5] END logistic__C=0.52, logistic__max_iter=10000,
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time=   0.1s
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total time=   0.0s
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total time=   0.1s
[CV 3/5] END logistic__C=0.52, logistic__max_iter=10000,
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total time=   0.0s
[CV 4/5] END logistic__C=0.52, logistic__max_iter=10000,
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total time=   0.1s
[CV 5/5] END logistic__C=0.52, logistic__max_iter=10000,
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total time=   0.0s
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0.1s
[CV 2/5] END logistic__C=0.52, logistic__max_iter=10000,
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0.0s
[CV 3/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
```

```
0.1s
[CV 4/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.780 total time=
0.0s
[CV 5/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.1s
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0.0s
[CV 4/5] END logistic__C=0.52, logistic__max_iter=10000,
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[CV 5/5] END logistic__C=0.52, logistic__max_iter=10000,
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total time=   0.0s
[CV 2/5] END logistic__C=0.52, logistic__max_iter=10000,
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[CV 3/5] END logistic__C=0.52, logistic__max_iter=10000,
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total time=   0.0s
[CV 4/5] END logistic__C=0.52, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.1s
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[CV 5/5] END logistic__C=0.52, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.0s
[CV 5/5] END logistic__C=0.52, logistic__max_iter=10000,
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total time=   0.0s
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score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.52, logistic__max_iter=10000,
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[CV 1/5] END logistic__C=0.52, logistic__max_iter=10000,
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time=   0.1s
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logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
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[CV 5/5] END logistic__C=0.54, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.54, logistic__max_iter=10000,
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time=   0.0s
[CV 2/5] END logistic__C=0.54, logistic__max_iter=10000,
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```

```
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[CV 2/5] END logistic__C=0.54, logistic__max_iter=10000,
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```

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0.1s
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0.0s
[CV 4/5] END logistic__C=0.54, logistic__max_iter=10000,
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[CV 3/5] END logistic__C=0.54, logistic__max_iter=10000,
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[CV 4/5] END logistic__C=0.54, logistic__max_iter=10000,
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time=   0.1s
[CV 4/5] END logistic__C=0.54, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.780 total
time=   0.1s
[CV 5/5] END logistic__C=0.54, logistic__max_iter=10000,
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total time=   0.0s
[CV 3/5] END logistic__C=0.54, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.792
total time=   0.1s
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total time=   0.0s
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0.0s

[CV 4/5] END logistic__C=0.54, logistic__max_iter=10000,
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0.1s
[CV 3/5] END logistic__C=0.54, logistic__max_iter=10000,
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0.0s
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logistic__penalty=l2, logistic__solver=saga;, score=0.780 total time=
0.0s
[CV 5/5] END logistic__C=0.54, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
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```

```
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time=   0.0s
[CV 4/5] END logistic__C=0.54, logistic__max_iter=10000,
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time=   0.1s
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score=nan total time=   0.0s
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```
0.0s
[CV 4/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.762 total time=
0.0s
[CV 5/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.930 total
time=   0.1s
[CV 2/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.826 total
time=   0.1s
[CV 4/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.762 total
time=   0.0s
[CV 5/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 1/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.905 total
time=   0.0s
[CV 2/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.1s
[CV 4/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.780 total
time=   0.1s
```

```
[CV 5/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
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[CV 1/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.905
total time=   0.1s
[CV 2/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.769
total time=   0.0s
[CV 3/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.792
total time=   0.1s
[CV 4/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.780
total time=   0.0s
[CV 5/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.1s

[CV 1/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
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[CV 3/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.1s
[CV 5/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.1s
[CV 2/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
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total time=   0.0s
[CV 1/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
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total time=   0.0s
[CV 2/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
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total time=   0.1s
[CV 3/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
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total time=   0.0s
[CV 5/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
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[CV 3/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
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score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.1s
[CV 2/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.1s
[CV 5/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.1s
[CV 3/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
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time=   0.0s
[CV 5/5] END logistic__C=0.5800000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.1s
[CV 1/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.1s
```

```
[CV 4/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.909 total
time=   0.1s
[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
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time=   0.1s
[CV 4/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.762 total
time=   0.1s
[CV 5/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 1/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.1s
[CV 3/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.1s
[CV 5/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.1s
[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
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total time=   0.0s

[CV 3/5] END logistic__C=0.6, logistic__max_iter=10000,
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[CV 4/5] END logistic__C=0.6, logistic__max_iter=10000,
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total time=   0.0s
[CV 5/5] END logistic__C=0.6, logistic__max_iter=10000,
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total time=   0.1s
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0.0s
[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.1s
[CV 3/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.1s
[CV 5/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.909 total time=
0.1s
[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.769 total time=
0.1s
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0.0s
[CV 4/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.762 total time=
0.1s
[CV 5/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.780 total time=
0.0s
[CV 5/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.6, logistic__max_iter=10000,
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```

```
time=   0.1s
[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
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time=   0.0s
[CV 3/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.826 total
time=   0.1s
[CV 4/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.762 total
time=   0.0s
[CV 5/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.905 total
time=   0.1s
[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
time=   0.0s
[CV 3/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.1s
[CV 4/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.780 total
time=   0.0s
[CV 5/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.1s
[CV 1/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.905
total time=   0.0s
[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.769
total time=   0.1s
[CV 3/5] END logistic__C=0.6, logistic__max_iter=10000,
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total time=   0.0s
[CV 4/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.780
total time=   0.1s
[CV 5/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.0s
[CV 1/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.905 total time=
0.1s
[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.769 total time=
0.0s
```

```
[CV 3/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.780 total time=
0.0s
[CV 5/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.905 total time=
0.0s
[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.780 total time=
0.0s
[CV 5/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
0.1s
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[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
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[CV 3/5] END logistic__C=0.6, logistic__max_iter=10000,
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time=   0.0s
[CV 4/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
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time=   0.1s
[CV 1/5] END logistic__C=0.6, logistic__max_iter=10000,
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total time=   0.0s
[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.6, logistic__max_iter=10000,
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logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.6, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
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[CV 3/5] END logistic__C=0.6, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.6, logistic__max_iter=10000,
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[CV 5/5] END logistic__C=0.62, logistic__max_iter=10000,
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time=   0.0s
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logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 3/5] END logistic__C=0.62, logistic__max_iter=10000,
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time=   0.1s
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logistic__penalty=l1, logistic__solver=liblinear;, score=0.762 total
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[CV 5/5] END logistic__C=0.62, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 1/5] END logistic__C=0.62, logistic__max_iter=10000,
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0.1s
[CV 2/5] END logistic__C=0.62, logistic__max_iter=10000,
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0.1s
[CV 5/5] END logistic__C=0.62, logistic__max_iter=10000,
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0.1s
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0.1s
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0.1s
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0.0s
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0.1s
[CV 5/5] END logistic__C=0.62, logistic__max_iter=10000,
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[CV 5/5] END logistic__C=0.62, logistic__max_iter=10000,
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time=   0.1s
[CV 1/5] END logistic__C=0.62, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.930 total
time=   0.1s
[CV 2/5] END logistic__C=0.62, logistic__max_iter=10000,
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total time=   0.1s
[CV 2/5] END logistic__C=0.62, logistic__max_iter=10000,
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total time=   0.1s
[CV 3/5] END logistic__C=0.62, logistic__max_iter=10000,
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total time=   0.1s
[CV 4/5] END logistic__C=0.62, logistic__max_iter=10000,
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total time=   0.1s
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total time=   0.1s
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[CV 3/5] END logistic__C=0.62, logistic__max_iter=10000,
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```

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[CV 5/5] END logistic__C=0.64, logistic__max_iter=10000,
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0.1s
[CV 2/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.64, logistic__max_iter=10000,
```

```
logistic__penalty=l2, logistic__solver=liblinear;, score=0.930 total
time=   0.1s
[CV 2/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.826 total
time=   0.0s
[CV 4/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.762 total
time=   0.1s
[CV 5/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 1/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.930 total
time=   0.0s
[CV 2/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.1s
[CV 4/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.780 total
time=   0.0s
[CV 5/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.1s
[CV 1/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.930
total time=   0.1s
[CV 2/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.769
total time=   0.0s
[CV 3/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.792
total time=   0.1s
[CV 4/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.780
total time=   0.1s
[CV 5/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.0s
[CV 1/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.930 total time=
0.1s
[CV 2/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.769 total time=
```

```
0.1s
[CV 3/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.780 total time=
0.0s
[CV 5/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.930 total time=
0.1s
[CV 2/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.780 total time=
0.0s
[CV 5/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.64, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.64, logistic__max_iter=10000,
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[CV 3/5] END logistic__C=0.64, logistic__max_iter=10000,
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time=   0.0s
[CV 4/5] END logistic__C=0.64, logistic__max_iter=10000,
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time=   0.0s
[CV 5/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.64, logistic__max_iter=10000,
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total time=   0.1s
[CV 2/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
```

```
[CV 4/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.1s
[CV 5/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.64, logistic__max_iter=10000,
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total time=   0.0s
[CV 3/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.1s
[CV 5/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.64, logistic__max_iter=10000,
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score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.64, logistic__max_iter=10000,
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score=nan total time=   0.1s
[CV 5/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.64, logistic__max_iter=10000,
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time=   0.0s
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time=   0.0s
[CV 4/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.64, logistic__max_iter=10000,
```

```
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s

[CV 2/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.1s
[CV 4/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.64, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
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logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
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logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.909 total
time=   0.1s
[CV 2/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 3/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.884 total
time=   0.0s
[CV 4/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.762 total
time=   0.0s
[CV 5/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
```

```
time=   0.0s
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logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.1s
[CV 3/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
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logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.909 total time=
0.1s
[CV 2/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.769 total time=
0.1s
```

```
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logistic__penalty=l1, logistic__solver=saga;, score=0.762 total time=
0.0s
[CV 5/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.930 total time=
0.1s
[CV 2/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.780 total time=
0.0s
[CV 5/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.930 total
time=   0.0s
[CV 2/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.0s
[CV 3/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.826 total
time=   0.1s
[CV 4/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.762 total
time=   0.0s
[CV 5/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.930 total
time=   0.0s
[CV 2/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.0s
[CV 4/5] END logistic__C=0.66, logistic__max_iter=10000,
```

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logistic__penalty=l2, logistic__solver=newton-cg;, score=0.780 total
time=   0.1s
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logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.0s
[CV 1/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.930
total time=   0.0s
[CV 2/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.769
total time=   0.1s
[CV 3/5] END logistic__C=0.66, logistic__max_iter=10000,
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total time=   0.0s
[CV 4/5] END logistic__C=0.66, logistic__max_iter=10000,
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total time=   0.0s
[CV 5/5] END logistic__C=0.66, logistic__max_iter=10000,
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total time=   0.1s
[CV 1/5] END logistic__C=0.66, logistic__max_iter=10000,
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0.0s
[CV 3/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.780 total time=
0.0s
[CV 5/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.930 total time=
0.1s
[CV 2/5] END logistic__C=0.66, logistic__max_iter=10000,
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0.0s
[CV 3/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.792 total time=
0.0s

[CV 4/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
```

```
0.0s
[CV 1/5] END logistic__C=0.66, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.66, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.0s
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[CV 4/5] END logistic__C=0.66, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.0s
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total time=   0.0s
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total time=   0.1s
[CV 3/5] END logistic__C=0.66, logistic__max_iter=10000,
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total time=   0.0s
[CV 4/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.66, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.66, logistic__max_iter=10000,
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score=nan total time=   0.1s
```

```
[CV 2/5] END logistic__C=0.66, logistic__max_iter=10000,
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logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
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0.1s
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0.0s
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logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.930 total
time=   0.0s
[CV 2/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.826 total
time=   0.0s
[CV 4/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.762 total
time=   0.1s
[CV 5/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 1/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.930 total
time=   0.1s
[CV 2/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.0s
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[CV 4/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.780 total
time=   0.1s
[CV 5/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.1s
[CV 1/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.930
total time=   0.1s
[CV 2/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.769
total time=   0.1s
[CV 3/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.792
total time=   0.0s
[CV 4/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
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total time=   0.1s
[CV 5/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.0s
[CV 1/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.930 total time=
0.1s
[CV 2/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.780 total time=
0.0s
[CV 5/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.930 total time=
0.1s
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[CV 3/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
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[CV 4/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.780 total time=
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[CV 5/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.0s
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[CV 1/5] END logistic__C=0.7000000000000001, logistic__max_iter=10000,
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time=   0.1s
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time=   0.1s
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logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
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[CV 3/5] END logistic__C=0.7200000000000001, logistic__max_iter=10000,
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logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
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time=   0.1s
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0.0s
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[CV 5/5] END logistic__C=0.7200000000000001, logistic__max_iter=10000,
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time=   0.1s
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time=   0.1s
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time=   0.1s
[CV 3/5] END logistic__C=0.7200000000000001, logistic__max_iter=10000,
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time=   0.1s
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time=   0.1s
[CV 5/5] END logistic__C=0.7200000000000001, logistic__max_iter=10000,
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total time=   0.1s
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0.1s
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[CV 2/5] END logistic__C=0.7200000000000001, logistic__max_iter=10000,
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[CV 3/5] END logistic__C=0.7200000000000001, logistic__max_iter=10000,
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[CV 4/5] END logistic__C=0.7200000000000001, logistic__max_iter=10000,
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time=   0.1s
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time=   0.1s
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time=   0.1s
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[CV 3/5] END logistic__C=0.74, logistic__max_iter=10000,
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logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
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time=   0.1s
[CV 2/5] END logistic__C=0.74, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
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time=   0.1s
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logistic__penalty=l1, logistic__solver=liblinear;, score=0.762 total
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time=   0.1s
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0.1s
[CV 5/5] END logistic__C=0.74, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.818 total time=
0.1s
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0.1s
[CV 3/5] END logistic__C=0.74, logistic__max_iter=10000,
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0.1s
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[CV 5/5] END logistic__C=0.74, logistic__max_iter=10000,
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time=   0.0s
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total time=   0.1s
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0.0s
[CV 5/5] END logistic__C=0.76, logistic__max_iter=10000,
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logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.76, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.930 total
time=   0.0s
[CV 2/5] END logistic__C=0.76, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.0s
[CV 3/5] END logistic__C=0.76, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.826 total
time=   0.1s
[CV 4/5] END logistic__C=0.76, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.762 total
time=   0.0s
[CV 5/5] END logistic__C=0.76, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
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logistic__penalty=l2, logistic__solver=newton-cg;, score=0.909 total
time=   0.1s
[CV 2/5] END logistic__C=0.76, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
time=   0.0s
[CV 3/5] END logistic__C=0.76, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.0s
[CV 4/5] END logistic__C=0.76, logistic__max_iter=10000,
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time=   0.1s
[CV 5/5] END logistic__C=0.76, logistic__max_iter=10000,
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time=   0.1s
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total time=   0.1s
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total time=   0.0s
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total time=   0.1s
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total time=   0.1s
[CV 5/5] END logistic__C=0.76, logistic__max_iter=10000,
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total time=   0.1s
[CV 1/5] END logistic__C=0.76, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.909 total time=
```

```
0.1s
[CV 2/5] END logistic__C=0.76, logistic__max_iter=10000,
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0.1s
[CV 3/5] END logistic__C=0.76, logistic__max_iter=10000,
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[CV 4/5] END logistic__C=0.76, logistic__max_iter=10000,
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0.0s
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[CV 4/5] END logistic__C=0.76, logistic__max_iter=10000,
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0.1s
[CV 5/5] END logistic__C=0.76, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.76, logistic__max_iter=10000,
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total time=   0.0s
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```
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total time=   0.0s
[CV 4/5] END logistic__C=0.76, logistic__max_iter=10000,
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total time=   0.1s
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total time=   0.1s
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[CV 4/5] END logistic__C=0.76, logistic__max_iter=10000,
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0.0s
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logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.78, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.909 total
time=   0.1s
[CV 2/5] END logistic__C=0.78, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
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[CV 4/5] END logistic__C=0.78, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.762 total
time=   0.0s
[CV 5/5] END logistic__C=0.78, logistic__max_iter=10000,
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```

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time=   0.1s
[CV 3/5] END logistic__C=0.78, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
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[CV 4/5] END logistic__C=0.78, logistic__max_iter=10000,
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[CV 1/5] END logistic__C=0.78, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.909 total time=
0.1s
```

```
[CV 2/5] END logistic__C=0.78, logistic__max_iter=10000,
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[CV 4/5] END logistic__C=0.78, logistic__max_iter=10000,
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0.1s
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logistic__penalty=l1, logistic__solver=saga;, score=0.818 total time=
0.1s
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0.1s
[CV 2/5] END logistic__C=0.78, logistic__max_iter=10000,
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0.0s
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0.1s
[CV 4/5] END logistic__C=0.78, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.78, logistic__max_iter=10000,
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time=   0.1s
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time=   0.0s
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[CV 5/5] END logistic__C=0.78, logistic__max_iter=10000,
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time=   0.1s
[CV 1/5] END logistic__C=0.78, logistic__max_iter=10000,
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time=   0.1s
[CV 2/5] END logistic__C=0.78, logistic__max_iter=10000,
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[CV 3/5] END logistic__C=0.78, logistic__max_iter=10000,
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time=   0.1s
[CV 5/5] END logistic__C=0.78, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.1s
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0.1s
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0.1s
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0.1s
[CV 4/5] END logistic__C=0.78, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.780 total time=
```

```
0.0s
[CV 5/5] END logistic__C=0.78, logistic__max_iter=10000,
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total time=   0.1s
[CV 5/5] END logistic__C=0.78, logistic__max_iter=10000,
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total time=   0.0s
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logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
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logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
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time=   0.1s
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time=   0.0s
[CV 4/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.762 total
time=   0.1s
[CV 5/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.909 total
time=   0.1s
[CV 2/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
time=   0.0s
```

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[CV 3/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.1s
[CV 4/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.780 total
time=   0.0s
[CV 5/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.1s
[CV 1/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.909
total time=   0.0s
[CV 2/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.769
total time=   0.1s
[CV 3/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.792
total time=   0.0s
[CV 4/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.780
total time=   0.0s
[CV 5/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.1s
[CV 1/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.909 total time=
0.0s
[CV 2/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.780 total time=
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[CV 5/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.909 total time=
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[CV 2/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
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logistic__penalty=l2, logistic__solver=saga;, score=0.780 total time=
0.1s

[CV 5/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
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time=   0.0s
[CV 4/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
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total time=   0.0s
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[CV 3/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.0s
[CV 5/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
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```

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score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
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score=nan total time=   0.1s
[CV 3/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
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score=nan total time=   0.1s
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time=   0.0s
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time=   0.0s
[CV 4/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.1s
[CV 5/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
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time=   0.0s
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time=   0.0s
[CV 4/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
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time=   0.0s
[CV 5/5] END logistic__C=0.8200000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.1s
[CV 1/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
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[CV 2/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
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logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.1s
[CV 1/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.909 total
time=   0.1s
[CV 2/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 3/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
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time=   0.0s
[CV 4/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.762 total
time=   0.1s
[CV 5/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
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time=   0.0s
[CV 2/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
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time=   0.0s
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time=   0.1s
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time=   0.0s
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logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.0s
[CV 3/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.0s
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0.0s
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logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
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0.0s

[CV 2/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.769 total time=
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0.1s
[CV 4/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.780 total time=
0.1s
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0.0s
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logistic__penalty=l2, logistic__solver=lbfgs;, score=0.909 total time=
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[CV 2/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
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0.1s
[CV 3/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.780 total time=
```

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0.1s
[CV 5/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
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time=   0.1s
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time=   0.1s
[CV 4/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
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[CV 4/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
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[CV 5/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
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logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.1s
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[CV 1/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.0s
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total time=   0.1s
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logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s

[CV 1/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
```

```
time=   0.0s
[CV 4/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
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[CV 5/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
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time=   0.1s
[CV 3/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.8400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.1s
[CV 1/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.1s
[CV 5/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.909 total
time=   0.1s
[CV 2/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 3/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.864 total
time=   0.1s
[CV 4/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.762 total
time=   0.0s
```

```
[CV 5/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.1s
[CV 5/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
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total time=   0.0s
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logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.1s
[CV 3/5] END logistic__C=0.86, logistic__max_iter=10000,
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total time=   0.0s
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[CV 4/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.86, logistic__max_iter=10000,
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logistic__penalty=l1, logistic__solver=saga;, score=0.884 total time=
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logistic__penalty=l1, logistic__solver=saga;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.844 total time=
0.0s
[CV 4/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.780 total time=
0.0s
[CV 5/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.800 total time=
0.0s
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logistic__penalty=l2, logistic__solver=lbfgs;, score=0.909 total time=
0.1s
[CV 2/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.930 total
time=   0.0s
[CV 2/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.826 total
time=   0.0s
[CV 4/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.762 total
time=   0.0s
[CV 5/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 1/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.909 total
time=   0.0s
[CV 2/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
```

```
time=   0.1s
[CV 3/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.1s
[CV 4/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.780 total
time=   0.1s
[CV 5/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.0s
[CV 1/5] END logistic__C=0.86, logistic__max_iter=10000,
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total time=   0.1s
[CV 2/5] END logistic__C=0.86, logistic__max_iter=10000,
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total time=   0.1s
[CV 3/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.792
total time=   0.0s
[CV 4/5] END logistic__C=0.86, logistic__max_iter=10000,
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total time=   0.0s

[CV 5/5] END logistic__C=0.86, logistic__max_iter=10000,
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total time=   0.1s
[CV 1/5] END logistic__C=0.86, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.86, logistic__max_iter=10000,
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0.0s
[CV 3/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.909 total time=
0.1s
[CV 2/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.792 total time=
0.1s
```

```
[CV 4/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
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time=   0.0s
[CV 2/5] END logistic__C=0.86, logistic__max_iter=10000,
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logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.86, logistic__max_iter=10000,
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time=   0.0s
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time=   0.0s
[CV 1/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
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total time=   0.1s
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total time=   0.0s
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logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.1s
[CV 3/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.86, logistic__max_iter=10000,
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total time=   0.0s
[CV 5/5] END logistic__C=0.86, logistic__max_iter=10000,
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score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.86, logistic__max_iter=10000,
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logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
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logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.1s
[CV 5/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
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logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.1s
[CV 3/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
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logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.1s
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time=   0.0s
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time=   0.0s
[CV 3/5] END logistic__C=0.86, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.1s
[CV 4/5] END logistic__C=0.86, logistic__max_iter=10000,
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time=   0.0s
[CV 5/5] END logistic__C=0.86, logistic__max_iter=10000,
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```

```
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[CV 3/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
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logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
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time=   0.0s
[CV 2/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
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logistic__penalty=l1, logistic__solver=liblinear;, score=0.864 total
time=   0.0s
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total time=   0.0s
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```
[CV 3/5] END logistic__C=0.88, logistic__max_iter=10000,
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logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
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logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
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logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
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logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.1s
[CV 1/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.884 total time=
0.0s
[CV 2/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.844 total time=
0.0s
[CV 4/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.800 total time=
0.0s
[CV 1/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.909 total time=
0.1s
[CV 2/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.88, logistic__max_iter=10000,
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```
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.780 total time=
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logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.930 total
time=   0.1s
[CV 2/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.826 total
time=   0.1s
[CV 4/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.762 total
time=   0.1s
[CV 5/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 1/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.909 total
time=   0.1s
[CV 2/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.1s
[CV 4/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.780 total
time=   0.1s
[CV 5/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.1s
[CV 1/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.909
total time=   0.1s
[CV 2/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.769
total time=   0.1s
[CV 3/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.792
total time=   0.1s
[CV 4/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.780
total time=   0.1s
[CV 5/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
```

```
total time=   0.2s
[CV 1/5] END logistic__C=0.88, logistic__max_iter=10000,
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0.1s
[CV 2/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.2s
[CV 1/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.909 total time=
0.1s
[CV 2/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.1s
[CV 3/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.1s
[CV 5/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.1s
```

```
[CV 2/5] END logistic__C=0.88, logistic__max_iter=10000,
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total time=   0.1s
[CV 3/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.88, logistic__max_iter=10000,
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total time=   0.1s
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total time=   0.1s
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[CV 4/5] END logistic__C=0.88, logistic__max_iter=10000,
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total time=   0.1s
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total time=   0.0s
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logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.1s
[CV 2/5] END logistic__C=0.88, logistic__max_iter=10000,
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[CV 4/5] END logistic__C=0.88, logistic__max_iter=10000,
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[CV 5/5] END logistic__C=0.88, logistic__max_iter=10000,
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score=nan total time=   0.0s

[CV 1/5] END logistic__C=0.88, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.88, logistic__max_iter=10000,
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logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.1s
[CV 4/5] END logistic__C=0.88, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.1s
[CV 5/5] END logistic__C=0.88, logistic__max_iter=10000,
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time=   0.0s
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time=   0.1s
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time=   0.0s
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0.1s
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logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
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[CV 5/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.1s
[CV 1/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.909 total
time=   0.1s
[CV 2/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 3/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.864 total
time=   0.1s
[CV 4/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.762 total
```

```
time=   0.0s
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time=   0.1s
[CV 2/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.1s
[CV 4/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.1s
[CV 1/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.9, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.0s
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total time=   0.1s
[CV 1/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.1s
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0.1s
[CV 5/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.1s
```

```
[CV 1/5] END logistic__C=0.9, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.9, logistic__max_iter=10000,
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0.1s
[CV 3/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.826 total time=
0.1s
[CV 4/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.800 total time=
0.1s
[CV 1/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.909 total time=
0.1s
[CV 2/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.1s
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logistic__penalty=l2, logistic__solver=liblinear;, score=0.930 total
time=   0.1s
[CV 2/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.826 total
time=   0.1s
[CV 4/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.762 total
time=   0.1s
[CV 5/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 1/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.909 total
time=   0.1s
[CV 2/5] END logistic__C=0.9, logistic__max_iter=10000,
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logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
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[CV 3/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.1s
[CV 4/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.780 total
time=   0.1s
[CV 5/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.1s
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total time=   0.0s
[CV 2/5] END logistic__C=0.9, logistic__max_iter=10000,
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total time=   0.1s
[CV 3/5] END logistic__C=0.9, logistic__max_iter=10000,
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total time=   0.1s
[CV 4/5] END logistic__C=0.9, logistic__max_iter=10000,
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total time=   0.1s
[CV 5/5] END logistic__C=0.9, logistic__max_iter=10000,
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total time=   0.1s
[CV 1/5] END logistic__C=0.9, logistic__max_iter=10000,
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0.0s
[CV 2/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.769 total time=
0.1s

[CV 3/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.909 total time=
0.1s
[CV 2/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.792 total time=
```

```
0.0s
[CV 4/5] END logistic__C=0.9, logistic__max_iter=10000,
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0.1s
[CV 5/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
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[CV 2/5] END logistic__C=0.9, logistic__max_iter=10000,
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time=   0.0s
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time=   0.1s
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[CV 2/5] END logistic__C=0.9, logistic__max_iter=10000,
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total time=   0.1s
[CV 3/5] END logistic__C=0.9, logistic__max_iter=10000,
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total time=   0.0s
[CV 4/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
```

```
[CV 5/5] END logistic__C=0.9, logistic__max_iter=10000,
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total time=   0.0s
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score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.9, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.1s
[CV 3/5] END logistic__C=0.9, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
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[CV 1/5] END logistic__C=0.92, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.92, logistic__max_iter=10000,
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logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
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[CV 5/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.909 total
time=   0.1s
[CV 2/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 3/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.864 total
time=   0.1s
[CV 4/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.762 total
time=   0.0s
[CV 5/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 1/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
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[CV 2/5] END logistic__C=0.92, logistic__max_iter=10000,
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time=   0.0s
[CV 4/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
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total time=   0.1s
[CV 2/5] END logistic__C=0.92, logistic__max_iter=10000,
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0.0s
[CV 1/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.884 total time=
0.1s
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logistic__penalty=l1, logistic__solver=saga;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.826 total time=
0.1s
[CV 4/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.780 total time=
0.0s
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logistic__penalty=l1, logistic__solver=saga;, score=0.800 total time=
0.0s
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0.1s
[CV 2/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.0s
```

```
[CV 4/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.930 total
time=   0.0s
[CV 2/5] END logistic__C=0.92, logistic__max_iter=10000,
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time=   0.0s
[CV 3/5] END logistic__C=0.92, logistic__max_iter=10000,
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time=   0.0s
[CV 4/5] END logistic__C=0.92, logistic__max_iter=10000,
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time=   0.0s
[CV 5/5] END logistic__C=0.92, logistic__max_iter=10000,
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time=   0.0s
[CV 1/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.909 total
time=   0.0s
[CV 2/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
time=   0.0s
[CV 3/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.0s
[CV 4/5] END logistic__C=0.92, logistic__max_iter=10000,
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time=   0.1s
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total time=   0.0s
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total time=   0.1s
[CV 3/5] END logistic__C=0.92, logistic__max_iter=10000,
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total time=   0.0s
[CV 4/5] END logistic__C=0.92, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.780
total time=   0.0s
[CV 5/5] END logistic__C=0.92, logistic__max_iter=10000,
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logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.1s
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0.1s
[CV 3/5] END logistic__C=0.92, logistic__max_iter=10000,
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0.0s
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[CV 5/5] END logistic__C=0.92, logistic__max_iter=10000,
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0.0s
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0.1s
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[CV 1/5] END logistic__C=0.92, logistic__max_iter=10000,
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```

```
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total time=   0.0s
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[CV 2/5] END logistic__C=0.92, logistic__max_iter=10000,
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```
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0.0s
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time=   0.1s
[CV 2/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
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logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
```

```
0.1s
[CV 1/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
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0.1s
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0.1s
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logistic__penalty=l1, logistic__solver=saga;, score=0.826 total time=
0.1s
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[CV 3/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
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0.1s
[CV 5/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
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time=   0.1s
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time=   0.1s
```

```
[CV 2/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
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[CV 3/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.1s
[CV 4/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.780 total
time=   0.0s
[CV 5/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.1s
[CV 1/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.909
total time=   0.1s
[CV 2/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.769
total time=   0.0s
[CV 3/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.792
total time=   0.1s
[CV 4/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.780
total time=   0.1s

[CV 5/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.0s
[CV 1/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.909 total time=
0.1s
[CV 2/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.909 total time=
0.1s
[CV 2/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
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logistic__penalty=l2, logistic__solver=saga;, score=0.792 total time=
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[CV 4/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
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time=   0.1s
[CV 2/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
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time=   0.0s
[CV 3/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
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time=   0.1s
[CV 4/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.1s
[CV 1/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.1s
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total time=   0.0s
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total time=   0.0s
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total time=   0.0s
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total time=   0.0s
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total time=   0.1s
[CV 4/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
```

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total time=   0.0s
[CV 5/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
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total time=   0.0s
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score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.1s
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score=nan total time=   0.0s
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score=nan total time=   0.1s
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[CV 2/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.1s
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time=   0.0s
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time=   0.0s
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time=   0.0s
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time=   0.0s
[CV 5/5] END logistic__C=0.9400000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.1s
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[CV 1/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
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0.0s
[CV 3/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
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[CV 4/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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0.0s
[CV 5/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 1/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.909 total
time=   0.1s
[CV 2/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 3/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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time=   0.0s
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time=   0.0s
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time=   0.0s
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time=   0.0s
[CV 2/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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[CV 4/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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time=   0.0s
[CV 5/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.1s
[CV 1/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
```

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[CV 2/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.0s
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total time=   0.1s
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total time=   0.0s
[CV 1/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
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logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
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0.1s
[CV 4/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
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logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
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[CV 2/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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0.1s
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0.1s
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0.1s
[CV 5/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.800 total time=
0.1s
[CV 1/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.909 total time=
0.1s
[CV 2/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
```

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0.1s
[CV 4/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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0.0s
[CV 5/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.1s
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time=   0.1s
[CV 3/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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time=   0.1s
[CV 1/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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time=   0.1s
[CV 3/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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time=   0.0s
[CV 4/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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time=   0.1s
[CV 5/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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time=   0.0s
[CV 1/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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total time=   0.1s
[CV 2/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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total time=   0.0s
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total time=   0.1s
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total time=   0.0s
[CV 5/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
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0.0s
[CV 2/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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0.1s
[CV 3/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
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[CV 4/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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[CV 1/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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[CV 2/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.769 total time=
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[CV 3/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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[CV 4/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.780 total time=
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time=   0.1s
[CV 3/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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time=   0.1s
[CV 5/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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time=   0.0s
[CV 1/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
```

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[CV 2/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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total time=   0.0s

[CV 4/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
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logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.1s
[CV 5/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.1s
[CV 2/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.9600000000000001, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.1s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.1s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.0s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=lbfgs;, score=nan total time=
0.1s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.909 total
time=   0.1s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.864 total
time=   0.1s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.780 total
```

```
time=   0.1s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.1s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.1s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cg;, score=nan total
time=   0.1s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.1s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=newton-cholesky;, score=nan
total time=   0.0s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.1s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.1s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=sag;, score=nan total time=
0.0s
```

```
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.884 total time=
0.1s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.826 total time=
0.1s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.780 total time=
0.1s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l1, logistic__solver=saga;, score=0.800 total time=
0.1s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.909 total time=
0.0s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.792 total time=
0.1s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.780 total time=
0.0s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=lbfgs;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.930 total
time=   0.0s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.769 total
time=   0.1s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.0s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.762 total
time=   0.0s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=liblinear;, score=0.800 total
time=   0.1s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.909 total
time=   0.1s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
```

```
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.769 total
time=   0.1s

[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.792 total
time=   0.1s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.780 total
time=   0.0s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cg;, score=0.818 total
time=   0.1s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.909
total time=   0.0s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.769
total time=   0.1s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.792
total time=   0.0s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.780
total time=   0.1s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=newton-cholesky;, score=0.818
total time=   0.0s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.909 total time=
0.0s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.769 total time=
0.1s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.792 total time=
0.0s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.780 total time=
0.0s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=sag;, score=0.818 total time=
0.1s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.909 total time=
0.0s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.769 total time=
0.0s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.792 total time=
```

```
0.1s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.780 total time=
0.0s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=l2, logistic__solver=saga;, score=0.818 total time=
0.0s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.1s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=lbfgs;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.1s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.0s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=liblinear;, score=nan
total time=   0.1s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.0s
```

```
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cg;, score=nan
total time=   0.1s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=newton-cholesky;,
score=nan total time=   0.0s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.1s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=sag;, score=nan total
time=   0.0s
[CV 1/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 2/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 3/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 4/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.0s
[CV 5/5] END logistic__C=0.98, logistic__max_iter=10000,
logistic__penalty=elasticnet, logistic__solver=saga;, score=nan total
time=   0.1s
[CV 1/5] END logistic__max_iter=10000, logistic__penalty=None,
```

```
logistic__solver=lbfgs;, score=0.889 total time=   0.1s
[CV 2/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=lbfgs;, score=0.769 total time=   0.1s
[CV 3/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=lbfgs;, score=0.864 total time=   0.1s
[CV 4/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=lbfgs;, score=0.762 total time=   0.1s
[CV 5/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=lbfgs;, score=0.791 total time=   0.1s
[CV 1/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 2/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 3/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 4/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=liblinear;, score=nan total time=   0.0s
[CV 5/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=liblinear;, score=nan total time=   0.1s
[CV 1/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=newton-cg;, score=0.889 total time=   0.1s
[CV 2/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=newton-cg;, score=0.769 total time=   0.1s
[CV 3/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=newton-cg;, score=0.864 total time=   0.1s

[CV 4/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=newton-cg;, score=0.762 total time=   0.1s
[CV 5/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=newton-cg;, score=0.791 total time=   0.1s
[CV 1/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=newton-cholesky;, score=0.889 total time=   0.0s
[CV 2/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=newton-cholesky;, score=0.769 total time=   0.0s
[CV 3/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=newton-cholesky;, score=0.864 total time=   0.1s
[CV 4/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=newton-cholesky;, score=0.762 total time=   0.0s
[CV 5/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=newton-cholesky;, score=0.791 total time=   0.0s
[CV 1/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=sag;, score=0.889 total time=   0.2s
[CV 2/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=sag;, score=0.769 total time=   0.2s
[CV 3/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=sag;, score=0.864 total time=   0.2s
[CV 4/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=sag;, score=0.762 total time=   0.1s
[CV 5/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=sag;, score=0.791 total time=   0.2s
```

```
[CV 1/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=saga;, score=0.889 total time=   0.2s
[CV 2/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=saga;, score=0.769 total time=   0.2s
[CV 3/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=saga;, score=0.864 total time=   0.2s
[CV 4/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=saga;, score=0.762 total time=   0.1s
[CV 5/5] END logistic__max_iter=10000, logistic__penalty=None,
logistic__solver=saga;, score=0.791 total time=   0.2s

/home/kf/.local/lib/python3.10/site-packages/sklearn/model_selection/
_validation.py:425: FitFailedWarning:
2455 fits failed out of a total of 4440.
The score on these train-test partitions for these parameters will be
set to nan.
If these failures are not expected, you can try to debug them by
setting error_score='raise'.

Below are more details about the failures:
----------------------------------------------------------------------
----------
245 fits failed with the following error:
Traceback (most recent call last):
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/model_selection/
_validation.py", line 732, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/pipeline.py",
line 420, in fit
    self._final_estimator.fit(Xt, y, **fit_params_last_step)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 1168, in fit
    solver = _check_solver(self.solver, self.penalty, self.dual)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 56, in _check_solver
    raise ValueError(
ValueError: Solver lbfgs supports only 'l2' or 'none' penalties, got
l1 penalty.

----------------------------------------------------------------------
```

```
----------
245 fits failed with the following error:
Traceback (most recent call last):
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/model_selection/
_validation.py", line 732, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/pipeline.py",
line 420, in fit
    self._final_estimator.fit(Xt, y, **fit_params_last_step)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 1168, in fit
    solver = _check_solver(self.solver, self.penalty, self.dual)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 56, in _check_solver
    raise ValueError(
ValueError: Solver newton-cg supports only 'l2' or 'none' penalties,
got l1 penalty.

----------------------------------------------------------------------
----------
245 fits failed with the following error:
Traceback (most recent call last):
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/model_selection/
_validation.py", line 732, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/pipeline.py",
line 420, in fit
    self._final_estimator.fit(Xt, y, **fit_params_last_step)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 1168, in fit
```

```
    solver = _check_solver(self.solver, self.penalty, self.dual)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 56, in _check_solver
    raise ValueError(
ValueError: Solver newton-cholesky supports only 'l2' or 'none'
penalties, got l1 penalty.

----------------------------------------------------------------------
----------
245 fits failed with the following error:
Traceback (most recent call last):
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/model_selection/
_validation.py", line 732, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/pipeline.py",
line 420, in fit
    self._final_estimator.fit(Xt, y, **fit_params_last_step)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 1168, in fit
    solver = _check_solver(self.solver, self.penalty, self.dual)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 56, in _check_solver
    raise ValueError(
ValueError: Solver sag supports only 'l2' or 'none' penalties, got l1
penalty.

----------------------------------------------------------------------
----------
245 fits failed with the following error:
Traceback (most recent call last):
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/model_selection/
_validation.py", line 732, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
```

```
"/home/kf/.local/lib/python3.10/site-packages/sklearn/pipeline.py",
line 420, in fit
    self._final_estimator.fit(Xt, y, **fit_params_last_step)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 1168, in fit
    solver = _check_solver(self.solver, self.penalty, self.dual)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 56, in _check_solver
    raise ValueError(
ValueError: Solver lbfgs supports only 'l2' or 'none' penalties, got
elasticnet penalty.

----------------------------------------------------------------------
----------
245 fits failed with the following error:
Traceback (most recent call last):
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/model_selection/
_validation.py", line 732, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/pipeline.py",
line 420, in fit
    self._final_estimator.fit(Xt, y, **fit_params_last_step)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 1168, in fit
    solver = _check_solver(self.solver, self.penalty, self.dual)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 66, in _check_solver
    raise ValueError(
ValueError: Only 'saga' solver supports elasticnet penalty, got
solver=liblinear.

----------------------------------------------------------------------
----------
245 fits failed with the following error:
```

```
Traceback (most recent call last):
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/model_selection/
_validation.py", line 732, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/pipeline.py",
line 420, in fit
    self._final_estimator.fit(Xt, y, **fit_params_last_step)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 1168, in fit
    solver = _check_solver(self.solver, self.penalty, self.dual)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 56, in _check_solver
    raise ValueError(
ValueError: Solver newton-cg supports only 'l2' or 'none' penalties,
got elasticnet penalty.

----------------------------------------------------------------------
----------
245 fits failed with the following error:
Traceback (most recent call last):
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/model_selection/
_validation.py", line 732, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/pipeline.py",
line 420, in fit
    self._final_estimator.fit(Xt, y, **fit_params_last_step)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 1168, in fit
    solver = _check_solver(self.solver, self.penalty, self.dual)
  File
```

```
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 56, in _check_solver
    raise ValueError(
ValueError: Solver newton-cholesky supports only 'l2' or 'none'
penalties, got elasticnet penalty.

--------------------------------------------------------------------
----------
245 fits failed with the following error:
Traceback (most recent call last):
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/model_selection/
_validation.py", line 732, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/pipeline.py",
line 420, in fit
    self._final_estimator.fit(Xt, y, **fit_params_last_step)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 1168, in fit
    solver = _check_solver(self.solver, self.penalty, self.dual)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 56, in _check_solver
    raise ValueError(
ValueError: Solver sag supports only 'l2' or 'none' penalties, got
elasticnet penalty.

--------------------------------------------------------------------
----------
245 fits failed with the following error:
Traceback (most recent call last):
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/model_selection/
_validation.py", line 732, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/pipeline.py",
line 420, in fit
```

```
    self._final_estimator.fit(Xt, y, **fit_params_last_step)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 1178, in fit
    raise ValueError("l1_ratio must be specified when penalty is
elasticnet.")
ValueError: l1_ratio must be specified when penalty is elasticnet.

--------------------------------------------------------------------------
----------
5 fits failed with the following error:
Traceback (most recent call last):
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/model_selection/
_validation.py", line 732, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/pipeline.py",
line 420, in fit
    self._final_estimator.fit(Xt, y, **fit_params_last_step)
  File "/home/kf/.local/lib/python3.10/site-packages/sklearn/base.py",
line 1151, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/linear_model/
_logistic.py", line 1227, in fit
    self.coef_, self.intercept_, self.n_iter_ = _fit_liblinear(
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/svm/_base.py",
line 1221, in _fit_liblinear
    solver_type = _get_liblinear_solver_type(multi_class, penalty,
loss, dual)
  File
"/home/kf/.local/lib/python3.10/site-packages/sklearn/svm/_base.py",
line 1060, in _get_liblinear_solver_type
    raise ValueError(
ValueError: Unsupported set of arguments: The combination of
penalty='None' and loss='logistic_regression' is not supported,
Parameters: penalty=None, loss='logistic_regression', dual=False

  warnings.warn(some_fits_failed_message, FitFailedWarning)
/home/kf/.local/lib/python3.10/site-packages/sklearn/model_selection/
_search.py:976: UserWarning: One or more of the test scores are non-
```

```
finite: [       nan 0.                nan        nan        nan 0.
 0.79536886 0.77254928 0.79536886 0.79536886 0.79536886 0.79536886
        nan        nan        nan        nan        nan        nan
        nan 0.                nan        nan        nan 0.
 0.80831768 0.79463812 0.80831768 0.80831768 0.80831768 0.80831768
        nan        nan        nan        nan        nan        nan
        nan 0.75253305        nan        nan        nan 0.648262
 0.80831768 0.80526628 0.80831768 0.80831768 0.80831768 0.80831768
        nan        nan        nan        nan        nan        nan
        nan 0.76675349        nan        nan        nan 0.70329373
 0.81345952 0.80713716 0.81345952 0.81345952 0.81345952 0.81345952
        nan        nan        nan        nan        nan        nan
        nan 0.79784562        nan        nan        nan 0.73573356
 0.81345952 0.80308858 0.81345952 0.81345952 0.81345952 0.81345952
        nan        nan        nan        nan        nan        nan
        nan 0.80465501        nan        nan        nan 0.79315858
 0.81345952 0.80308858 0.81345952 0.81345952 0.81345952 0.81345952
        nan        nan        nan        nan        nan        nan
        nan 0.7989547         nan        nan        nan 0.80655978
 0.81895635 0.80731691 0.81895635 0.81895635 0.81895635 0.81895635
        nan        nan        nan        nan        nan        nan
        nan 0.80281018        nan        nan        nan 0.80970826
 0.81895635 0.81277145 0.81895635 0.81895635 0.81895635 0.81895635
        nan        nan        nan        nan        nan        nan
        nan 0.80586836        nan        nan        nan 0.81520508
 0.81895635 0.81277145 0.81895635 0.81895635 0.81895635 0.81895635
        nan        nan        nan        nan        nan        nan
        nan 0.81532669        nan        nan        nan 0.81607089
 0.82272158 0.81277145 0.82272158 0.82272158 0.82272158 0.82272158
        nan        nan        nan        nan        nan        nan
        nan 0.82834562        nan        nan        nan 0.81607089
 0.82272158 0.81277145 0.82272158 0.82272158 0.82272158 0.82272158
        nan        nan        nan        nan        nan        nan
        nan 0.82834562        nan        nan        nan 0.81607089
 0.82272158 0.81277145 0.82272158 0.82272158 0.82272158 0.82272158
        nan        nan        nan        nan        nan        nan
        nan 0.82834562        nan        nan        nan 0.81237061
 0.82272158 0.81277145 0.82272158 0.82272158 0.82272158 0.82272158
        nan        nan        nan        nan        nan        nan
        nan 0.82704088        nan        nan        nan 0.81237061
 0.82609037 0.81636553 0.82609037 0.82609037 0.82609037 0.82609037
        nan        nan        nan        nan        nan        nan
        nan 0.83126921        nan        nan        nan 0.81237061
 0.82204179 0.82122811 0.82204179 0.82204179 0.82204179 0.82204179
        nan        nan        nan        nan        nan        nan
        nan 0.83517165        nan        nan        nan 0.81237061
 0.82204179 0.82122811 0.82204179 0.82204179 0.82204179 0.82204179
        nan        nan        nan        nan        nan        nan
        nan 0.83517165        nan        nan        nan 0.81607089
```

```
 0.82204179 0.82122811 0.82204179 0.82204179 0.82204179 0.82204179
        nan        nan        nan        nan        nan        nan
        nan 0.82746506        nan        nan        nan 0.81607089
 0.82204179 0.82459691 0.82204179 0.82204179 0.82204179 0.82204179
        nan        nan        nan        nan        nan        nan
        nan 0.82746506        nan        nan        nan 0.81607089
 0.82204179 0.82459691 0.82204179 0.82204179 0.82204179 0.82204179
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.81607089
 0.81867299 0.82459691 0.81867299 0.81867299 0.81867299 0.81867299
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.81607089
 0.81867299 0.82459691 0.81867299 0.81867299 0.81867299 0.81867299
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.82205095
 0.81867299 0.82459691 0.81867299 0.81867299 0.81867299 0.81867299
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.82205095
 0.81867299 0.82459691 0.81867299 0.81867299 0.81867299 0.81867299
        nan        nan        nan        nan        nan        nan
        nan 0.83601732        nan        nan        nan 0.8255947
 0.81867299 0.81973433 0.81867299 0.81867299 0.81867299 0.81867299
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.8255947
 0.81286579 0.81397575 0.81286579 0.81286579 0.81286579 0.81286579
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.8255947
 0.81286579 0.81397575 0.81286579 0.81286579 0.81286579 0.81286579
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.8255947
 0.81286579 0.81397575 0.81286579 0.81286579 0.81286579 0.81286579
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.8255947
 0.81286579 0.81749101 0.81286579 0.81286579 0.81286579 0.81286579
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.8255947
 0.81286579 0.81749101 0.81286579 0.81286579 0.81286579 0.81286579
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.8255947
 0.81286579 0.81749101 0.81286579 0.81286579 0.81286579 0.81286579
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.8255947
 0.81795992 0.81749101 0.81795992 0.81795992 0.81795992 0.81795992
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.8255947
 0.81795992 0.81749101 0.81795992 0.81795992 0.81795992 0.81795992
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.82057054
 0.81795992 0.81749101 0.81795992 0.81795992 0.81795992 0.81795992
```

```
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.82440892
 0.81795992 0.81749101 0.81795992 0.81795992 0.81795992 0.81795992
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.82057054
 0.81795992 0.81749101 0.81795992 0.81795992 0.81795992 0.81795992
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.82057054
 0.81795992 0.81749101 0.81795992 0.81795992 0.81795992 0.81795992
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.82428715
 0.81795992 0.81749101 0.81795992 0.81795992 0.81795992 0.81795992
        nan        nan        nan        nan        nan        nan
        nan 0.83094332        nan        nan        nan 0.82428715
 0.81373159 0.81749101 0.81373159 0.81373159 0.81373159 0.81373159
        nan        nan        nan        nan        nan        nan
        nan 0.82692641        nan        nan        nan 0.82428715
 0.81373159 0.81749101 0.81373159 0.81373159 0.81373159 0.81373159
        nan        nan        nan        nan        nan        nan
        nan 0.82692641        nan        nan        nan 0.82065079
 0.81373159 0.81749101 0.81373159 0.81373159 0.81373159 0.81373159
        nan        nan        nan        nan        nan        nan
        nan 0.82692641        nan        nan        nan 0.82065079
 0.81373159 0.81749101 0.81373159 0.81373159 0.81373159 0.81373159
        nan        nan        nan        nan        nan        nan
        nan 0.82692641        nan        nan        nan 0.81557679
 0.81373159 0.81749101 0.81373159 0.81373159 0.81373159 0.81373159
        nan        nan        nan        nan        nan        nan
        nan 0.82692641        nan        nan        nan 0.81557679
 0.81373159 0.81749101 0.81373159 0.81373159 0.81373159 0.81373159
        nan        nan        nan        nan        nan        nan
        nan 0.82692641        nan        nan        nan 0.81557679
 0.81373159 0.81749101 0.81373159 0.81373159 0.81373159 0.81373159
        nan        nan        nan        nan        nan        nan
        nan 0.82692641        nan        nan        nan 0.81190529
 0.81373159 0.81749101 0.81373159 0.81373159 0.81373159 0.81373159
        nan        nan        nan        nan        nan        nan
        nan 0.82692641        nan        nan        nan 0.81190529
 0.81373159 0.81227362 0.81373159 0.81373159 0.81373159 0.81373159
        nan        nan        nan        nan        nan        nan
        nan 0.82692641        nan        nan        nan 0.81190529
 0.81373159 0.81227362 0.81373159 0.81373159 0.81373159 0.81373159
        nan        nan        nan        nan        nan        nan
        nan 0.83064302        nan        nan        nan 0.81190529
 0.81373159 0.81227362 0.81373159 0.81373159 0.81373159 0.81373159
        nan        nan        nan        nan        nan        nan
        nan 0.83064302        nan        nan        nan 0.81190529
 0.81373159 0.81227362 0.81373159 0.81373159 0.81373159 0.81373159
        nan        nan        nan        nan        nan        nan
```

```
 0.81487169        nan 0.81487169 0.81487169 0.81487169 0.81487169]
  warnings.warn(

GridSearchCV(cv=5,
             estimator=Pipeline(steps=[('preprocesor',

ColumnTransformer(transformers=[('num',

Pipeline(steps=[('imputer',

KNNImputer()),

('scaler',

MinMaxScaler())]),

['age',

'trestbps',

'chol',

'thalach',

'ca',

'oldpeak']),

('cat',

Pipeline(steps=[('imputer',

SimpleImputer(strategy='most_frequent')),

('encoder',

OneHotEncoder(drop='first'))]),

['cp',

'restecg...
       0.9 , 0.92, 0.94, 0.96, 0.98]),
                          'logistic__max_iter': [10000],
                          'logistic__penalty': ['l1', 'l2',
'elasticnet'],
                          'logistic__solver': ['lbfgs', 'liblinear',
                                               'newton-cg', 'newton-
cholesky',
                                               'sag', 'saga']},
                         {'logistic__max_iter': [10000],
```

```
                          'logistic__penalty': [None],
                          'logistic__solver': ['lbfgs', 'liblinear',
                                               'newton-cg', 'newton-
cholesky',
                                               'sag', 'saga']}],
             scoring='f1', verbose=4)
```

```
best_logistic = gscv_logistic.best_estimator_
best_hyperparameters = gscv_logistic.best_params_
```

```
best_hyperparameters
```

```
{'logistic__C': 0.48000000000000004,
 'logistic__max_iter': 10000,
 'logistic__penalty': 'l1',
 'logistic__solver': 'liblinear'}
```

```
y_pred = best_logistic.predict(X_test)
```

```
print(classification_report(y_pred, y_test))
```

```
              precision    recall  f1-score   support

           0       0.85      0.80      0.82        35
           1       0.75      0.81      0.78        26

    accuracy                           0.80        61
   macro avg       0.80      0.80      0.80        61
weighted avg       0.81      0.80      0.80        61
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Your implementation of logistic regression using a comprehensive preprocessor and grid search for hyperparameter tuning is well-structured. Here are a few comments and suggestions for improvement to ensure optimal model performance and code efficiency:

1. Preprocessing Setup Your preprocessing pipelines for different types of variables (numerical, categorical, ordinal, binary) are well-configured. Using KNNImputer for numerical data and SimpleImputer for categorical data is a good practice as it appropriately handles missing values based on the nature of the data. The use of MinMaxScaler for numerical data ensures that the magnitude of the variables does not affect the model, which is particularly important for logistic regression.

2. Pipeline Configuration Your pipeline integrates the preprocessing and logistic regression model seamlessly. This setup ensures that all transformations and model training occur within a single, contained workflow, reducing the risk of data leakage and simplifying the model deployment process.

3. Grid Search Setup Penalty and Solver Compatibility: You have included a variety of solvers and penalties in your hyperparameter grid. It's crucial to ensure compatibility between solvers and penalty terms. For example, liblinear works well

with l1 and l2, but not with elasticnet or None. Conversely, saga can handle l1, l2, elasticnet, and None. Parameter Range: You've set a fine step for C, which is great for a detailed search, but consider expanding the range of C values to see if the model benefits from more or less regularization, especially in non-linear decision boundaries. Solver Options: Note that "newton-cholesky" is not a standard solver option in LogisticRegression. The typical solvers are newton-cg, lbfgs, liblinear, sag, and saga.

4. Model Training and Evaluation Fitting the Model: Ensure that your training data (X_train, y_train) does not have any preprocessing steps applied outside of the pipeline to avoid data leakage. Model Evaluation: After training, using classification_report provides a comprehensive overview of the model's performance. Given logistic regression's output, you may also want to consider additional metrics such as ROC AUC for a more detailed analysis of model performance across different thresholds, which is useful for imbalanced datasets.

5. Code Improvements Solver Cleanup: Remove the non-existent solver "newton-cholesky" from your grid. Expansion of C: Consider increasing the range of C to beyond 1 (e.g., up to 10 or 100) to explore less regularized models, which might be beneficial depending on your dataset characteristics. Elastic Net: If using elasticnet, you need to specify l1_ratio as a parameter in your grid search. This parameter controls the mix between l1 and l2 regularization. Example Adjusted Grid Search Configuration python Copy code hyperparameters = [{ "logistic__penalty": ["l1", "l2", "elasticnet"], "logistic__C": np.arange(start_C, end_C, step_C), "logistic__l1_ratio": [0.5], # Only if elasticnet is used "logistic__max_iter": [10000], "logistic__solver": ["lbfgs", "liblinear", "sag", "saga"] # Corrected solver options }, { "logistic__penalty": [None], "logistic__max_iter": [10000], "logistic__solver": ["lbfgs", "sag", "saga"] # Appropriate solvers for no penalty }] Your setup for logistic regression is solid, but paying attention to these details can help optimize model training and ensure the accuracy of your results.